Problem Set 11
Econ 5253 – Spring 2026

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Question 6: Summary Statistics

The summary statistics make sense for a sample of working women in 1988. Mean years of schooling `hgc` is around the high-school-plus-some-college range, and `exper` averages a moderate number of years on the current job. Roughly 20% of the sample is in a union job, around 60% are married, and roughly half have at least one child living at home.

The missing rate of `logwage` is **30.7%**. Given the sample is restricted to women already known to be working, the missingness is most likely **Missing Not At Random (MNAR)** or at best Missing At Random (MAR): wages are most likely missing for women whose wage outcomes are systematically related to unobserved factors (e.g., women who work informally or report wages selectively). The selection problem is exactly what the Heckman model in Q7 is designed to address.

Question 7: Imputation Methods and Returns to Schooling

I estimate

$$\log wage_i = \beta_0 + \beta_1 hgc_i + \beta_2 union_i + \beta_3 college_i + \beta_4 exper_i + \beta_5 exper_i^2 + \eta_i \tag{1}$$

under three approaches to handling missing `logwage`: (i) listwise deletion (complete cases), (ii) mean imputation, and (iii) Heckman two-step selection.

The coefficient of interest β_1 (returns to schooling, true value 0.091) takes the following values:

Method	$\hat{\beta}_1$
Complete cases	0.0590
Mean imputation	0.0363
Heckman 2-step	0.0915
True value	0.0910

Discussion. Complete-cases regression underestimates the true returns to schooling at 0.059, because the missing observations are non-random and listwise deletion drops them in a way that

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
logwage	1546	31	1.7	0.7	-1.0	1.7	4.2	
hgc	14	0	12.5	2.4	5.0	12.0	18.0	
exper	1932	0	6.4	4.9	0.0	6.0	25.0	
kids	2	0	0.4	0.5	0.0	0.0	1.0	
married$_n um$	2	0	0.6	0.5	0.0	1.0	1.0	
kids$_n um$	2	0	0.4	0.5	0.0	0.0	1.0	
union$_n um$	2	0	0.2	0.4	0.0	0.0	1.0	
college$_n um$	2	0	0.1	0.3	0.0	0.0	1.0	
		N	college	0	1996	89.5		
	1	233	10.5					
married	0	814	36.5					
	1	1415	63.5					
union	0	1700	76.3					
	1	529	23.7					

biases the remaining sample. Mean imputation performs the worst at 0.036; filling in missing values with the unconditional mean compresses the variance of $\log wage$ and attenuates the slope on `hgc` toward zero. The Heckman two-step estimator recovers $\hat{\beta}_1 = 0.0915$, essentially the true value. By explicitly modeling the selection equation (the probability of observing a wage as a function of `married` and `kids`, in addition to the wage covariates), Heckman corrects for the non-random nature of the missingness and delivers a consistent estimate.

The pattern confirms the standard ordering of these methods under non-random missingness: mean imputation is the worst, complete-cases is biased but not catastrophic, and a properly-specified selection model is best when an exclusion restriction is available.

Question 8: Probit Model of Union Job Preference

The probit model

$$u_i = \alpha_0 + \alpha_1 hgc_i + \alpha_2 college_i + \alpha_3 exper_i + \alpha_4 married_i + \alpha_5 kids_i + \varepsilon_i$$

yields the following estimates:

Table 1: Imputation Methods: Returns to Schooling

	Complete Cases	Mean Imputation	Heckman
(Intercept)	0.834***	1.149***	0.446***
	(0.113)	(0.078)	(0.122)
hgc	0.059***	0.036***	0.091***
	(0.009)	(0.006)	(0.010)
union1	0.222*	0.068	0.186*
	(0.087)	(0.047)	(0.084)
college1	−0.065	−0.126**	0.092
	(0.106)	(0.048)	(0.100)
exper	0.050***	0.021**	0.054***
	(0.013)	(0.007)	(0.012)
I(exper^2)	−0.004**	−0.001**	−0.002+
	(0.001)	(0.000)	(0.001)
invMillsRatio			−0.695***
			(0.060)
Num.Obs.	1545	2229	2229
R2	0.038	0.020	
R2 Adj.	0.035	0.018	

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Variable	Estimate	Std. Error	z	p
(Intercept)	-6.743	0.804	-8.39	< 0.001
`hgc`	-1.009	0.098	-10.34	< 0.001
`college`	0.397	0.427	0.93	0.352
`exper`	1.849	0.156	11.86	< 0.001
`married`	0.588	0.206	2.86	0.004
`kids`	0.799	0.202	3.96	< 0.001

Question 9: Counterfactual Policy

I simulate a policy in which there is no preference or penalty to wives or mothers for working in union jobs by setting the coefficients on `married` and `kids` to zero and recomputing predicted probabilities. The results are:

	Avg. P(union)
Baseline	0.2373
Counterfactual ($\alpha_4 = \alpha_5 = 0$)	0.2174
Difference	-0.0199

Removing the marriage and kids effects lowers the average predicted probability of holding a union job by about 2 percentage points. This indicates that, in the estimated model, being married and having kids *raises* the probability of union employment, so removing those effects mechanically lowers it.

Is the model realistic? No. Several features of the estimates raise concerns:

1. The coefficient on `hgc` is -1.009, implying that one additional year of schooling *decreases* the latent utility of union employment by an enormous amount. This is implausibly large in magnitude and likely the wrong sign—empirically, union membership in the late 1980s was concentrated among workers with moderate education, not strongly negatively related to schooling at this magnitude.

2. The coefficient on `exper` is $+1.849$, which is also implausibly large in absolute value for a probit model.

3. R issued the warning "fitted probabilities numerically 0 or 1 occurred," which is a classic symptom of separation or near separation in the data and a signal that the model is numerically unstable.

4. The model omits demand-side variables (industry, occupation, geography) that drive union availability. With those omitted, the coefficients on individual characteristics absorb correlated omitted-variable effects.

The counterfactual is therefore best read as descriptive: it tells us how the average predicted probability would change if the `married` and `kids` coefficients were zeroed out *within this particular fitted model*, not as a credible estimate of how a real-world policy would change union participation. A more defensible model would include industry/occupation controls and would not produce coefficients of magnitude greater than one on continuous covariates.